

Difference-based Questions

→ ArrayList vs Linked List

ArrayList	Linked List
• ArrayList uses dynamic array to store elements in it. • Insertion operation is slow. • It implements a List interface. • Deletion operation is not very efficient. • It is used to store only similar types of data. • Less memory is used.	• Linked List uses double linked list to store the elements in it. • Insertion operation is fast. • It implements both the List interface and Deque. • Deletion operation is very efficient. • It is used to store any types of data. • More memory is used.

→ Stack vs Vector

Stack	Vector
• A subclass of Vector that implements LIFO. • We can access data by using stack methods. (push, pop, peek) • Stack is synchronized. • Stack extends Vector class. • Stack is used for Browser history.	• A dynamic array that grows automatically. • We can access data using get(index). • Vector is also synchronized. • Vector implements List interface. • Vector is also used for thread safe history.

→ ArrayList vs Vector

ArrayList	Vector
• Increased size by 50% when it is full • When working in single threaded programs • Not synchronized • Fast access but not thread safe	• Increased size by double when it is full. • When working in multi threaded programs. • Synchronized. • thread - safe ArrayList

Real life Scenario :

For a browser history feature, we can use stack.

stack having last in first out behaviour.

If you visit new URLs, they should be stored in order.

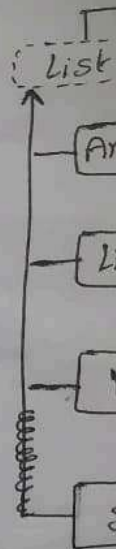
When you click back button it fetch url from stack and display last page.

To achieve browser history feature, we have to take forward and backstack.

forward stack is used to store pages when you press back, to go forward later.

backward stack is used to store visited pages.

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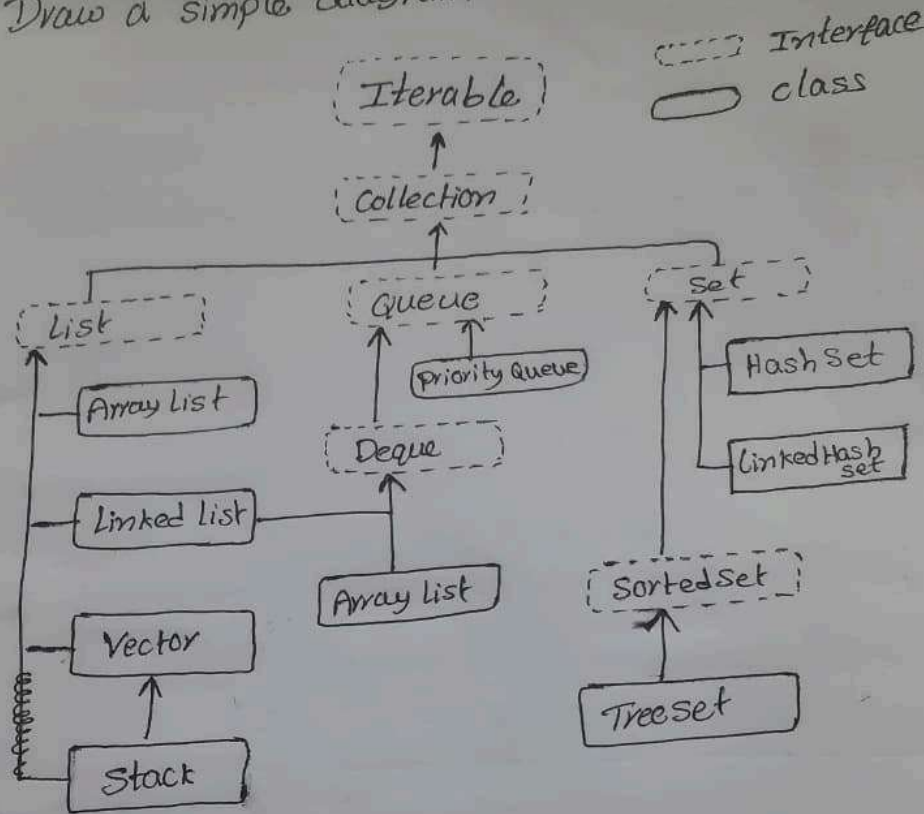
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Draw a simple diagrams



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